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31338 Network Servers
32520 Systems Administration

Week 4

Advanced Networking -- Dynamic Host Configuration Protocol (DHCP)

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Lab 3a Recapture

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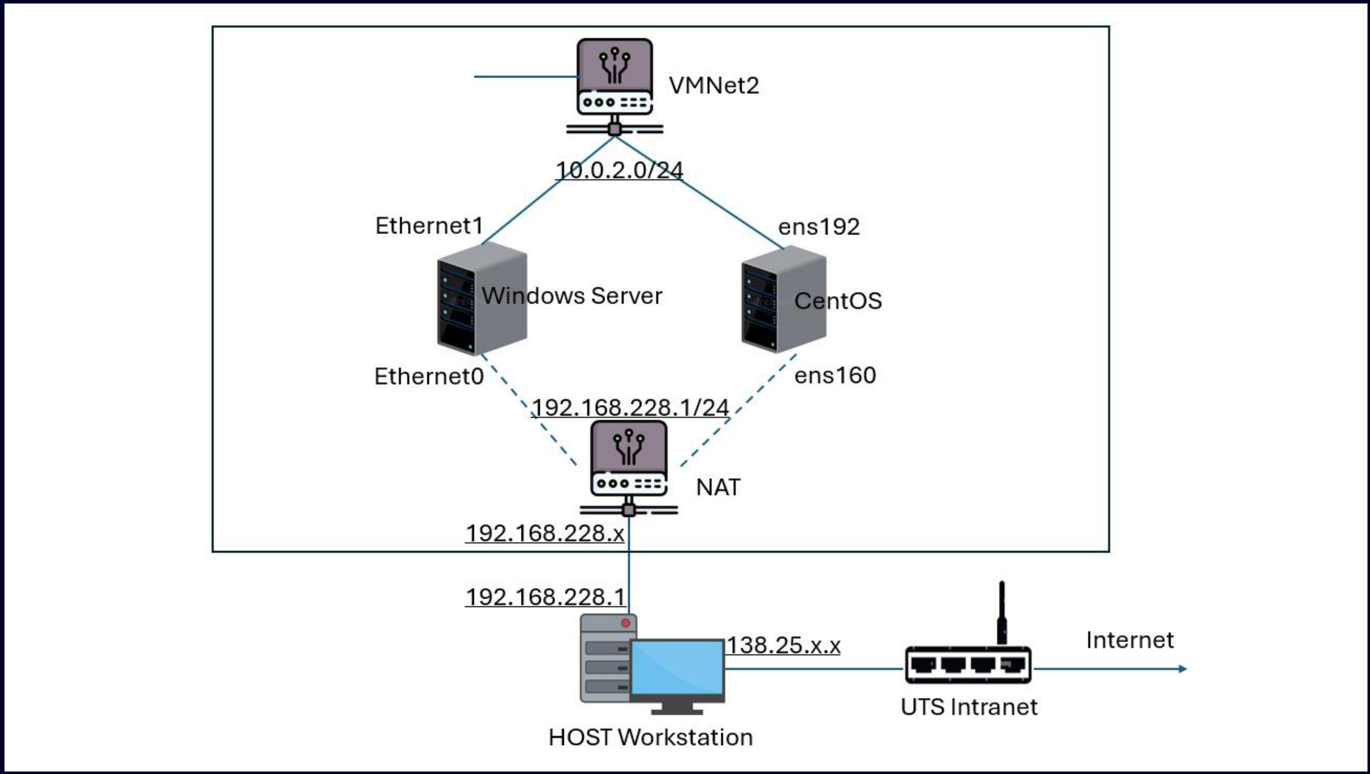
Each VM has two virtual interfaces: CentOS: ens160 & ens192
Windows: Ethenet0 & 1
They are v-NIC (Network Interface Card)

For Vmnet2, private network range: 10.0.2.0/24 – 10.0.2.254/24.

For NAT, private network portion: 192.168.228.0/24 – 192.168.228.254/24



Lab 3a Recapture

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- Static network configuration
  - Persistent setting in the file ens192.nmconnection
    - address1=10.0.2.1/24
    - gateway=10.0.2.1
  - Windows network config should go through the "Protocol Version 4 (TCP/IPv4) "
    - IP address=10.0.2.2
    - Network mask=255.255.255.0
    - gateway=10.0.2.1
  - Questions:
    - from the Linux VM to ping your host laptop or lab PC ?
    - from your host PC to ping ens160 and ens192?
- Setting up the firewall to let Linux to ping Windows

Lab 3 & Lab 4

- About nmconnection files
  - Auto-config after the reboot and keep it through nmtui
  - View ens160.nmconnection using the cat command
    - method=auto --the interface will be dynamically assigned an IP address by a DHCP server.
    - autoconnect=true --interface is enabled when Linux boots up
  - Edit ens192.nmconnection using a text editor
    - address1=10.0.2.1/24
    - gateway=10.0.2.1
    - method=manual
    - autoconnect=true
  - nmcli con <reload|down|up> <ens160|ens192>

More things to do

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- How to change hostname and, DNS and search domain
  - Change it using hostnamectl
    - hostnamectl set-hostname <hostname>
  - Inspect the new hostname
    - cat /etc/hostname
    - hostname
  - vim ens192.nmconnection
    - dns=1.1.1.1;2.2.2.2;
    - dns-search=it.uts.edu.au;google.com.au;

DHCP Introduction

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- Dynamic Host Configuration Protocol (DHCP) reduces network complexity and administration

- Static IP address & configuration
  - As per static network – non-DHCP client, manually configured
  - Question: How to cope with 10000 machines?

- DHCP clients:
  - Automatically obtain network config from a DHCP server.
  - Use UDP port 67/68
  - Uses network broadcasts on local subnet 255:255:255:255

Broadcast: **Whatever IP network a particular host is on, that host can always use this IP address to send a packet to every node on the Local Network. In our topology, Host 1 could send a message to the IP address 255.255. 255.255 to speak to everyone else on it's own local network.**

Address Allocation

- Dynamic IP configuration
  - Allocate "lease" IP address from a 'pool' of addresses
    - Can name the pool – called '**scopes**' in Windows
    - "**lease**" is for a specific adapter MAC & for specified length of time
    - Will change over time
  - DHCP can also "reserve" IP addresses for specific hosts
    - Good for servers, printers, hosts

Addresses Allocation

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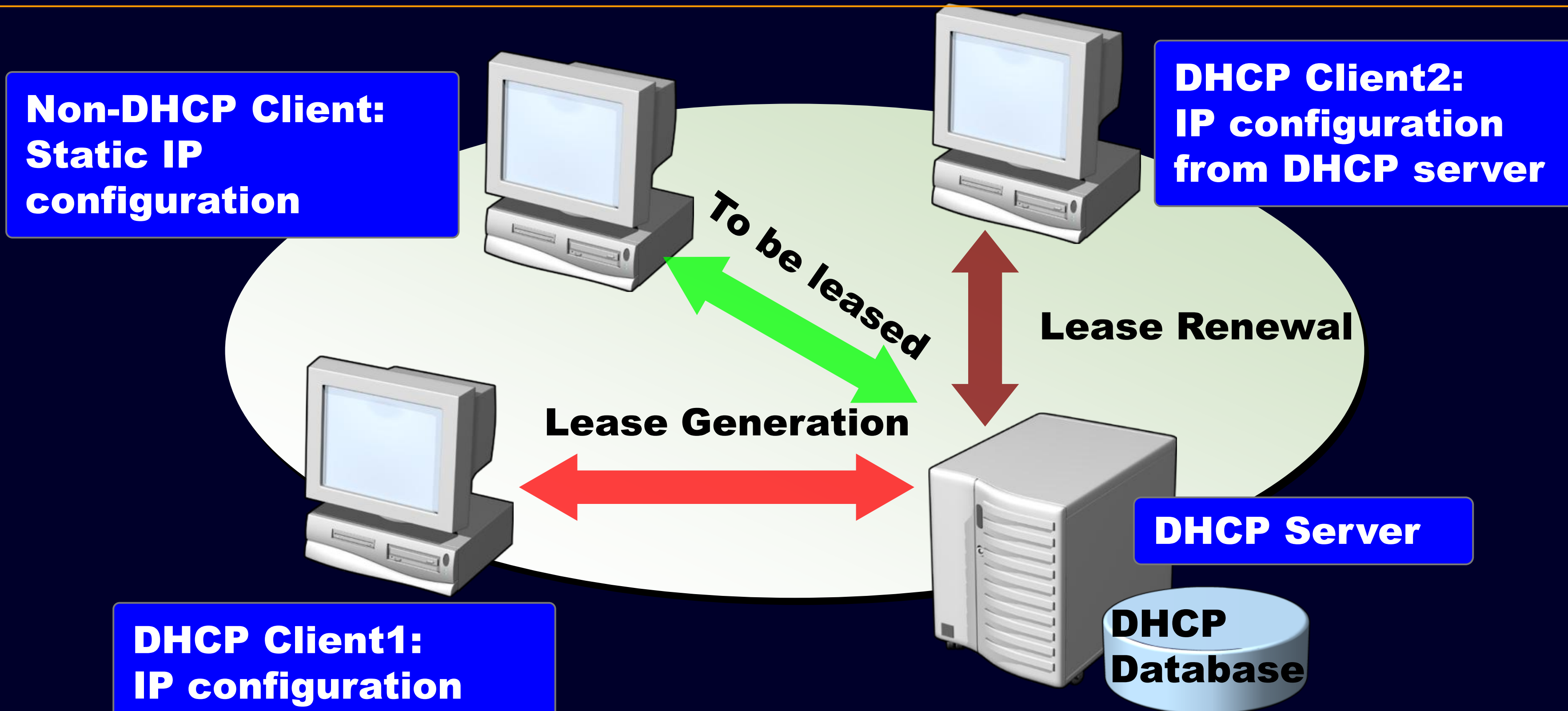
Two methods for obtaining a lease are to request a new lease or to renew an existing lease.



IP Address1: Leased to DHCP Client1 (Gen)
IP Address2: Leased to DHCP Client2 (Ren)
IP Address3: Available to be leased (To be)

DHCP server can be relayed through an agent

DHCP Operations

DHCP Client

- broadcasts DHCPDISCOVER UDP packet on local broadcasts subnet
- Chooses lease & DHCPREQUEST packet
- Can also reply with DHCPDECLINE if duplicate

DHCP Server

- Broadcasts DHCPOFFER with lease information
- Selected server replies with DHCPACK, DHCPNAK

For some reason, one agent will reply with DHCPRELEASE-- The agent will forward the message to a DHCP server which is configured

How DHCP Lease Generation Works

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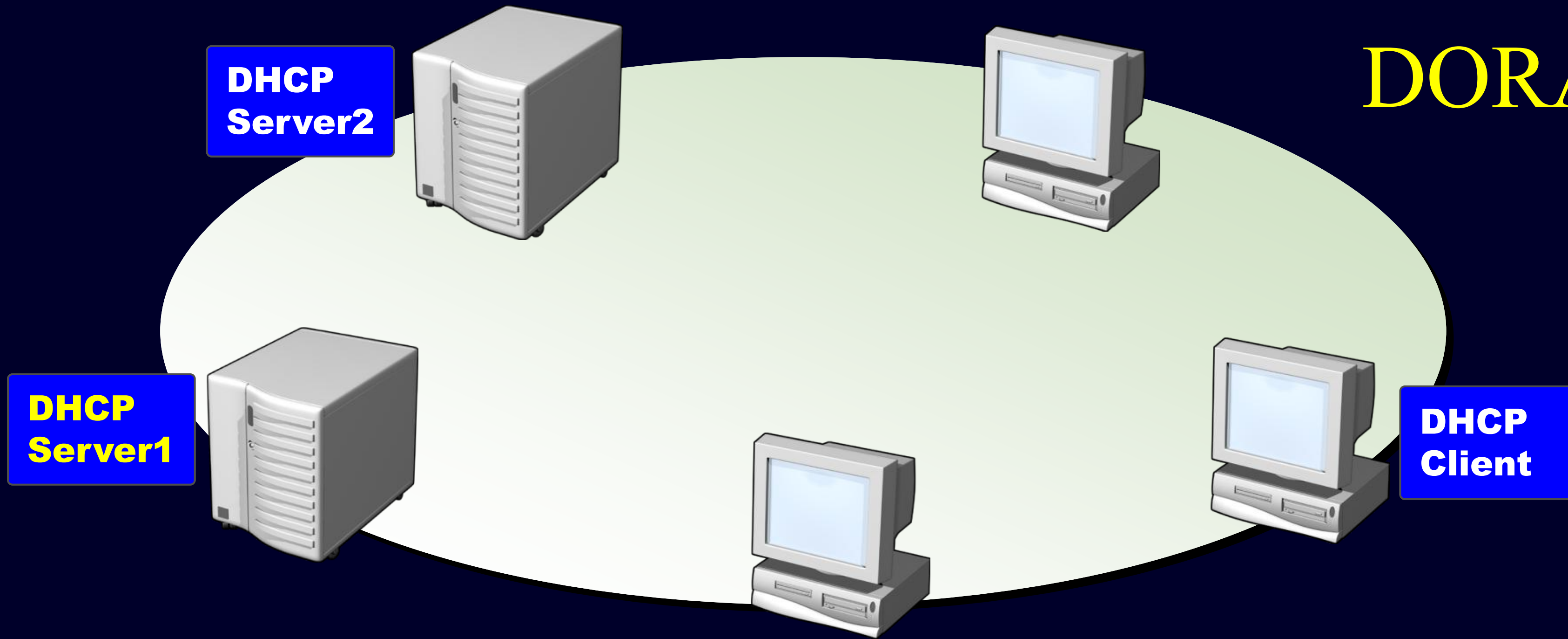
DORA



1. DHCP client broadcasts a DHCPDISCOVER packet
2. DHCP servers broadcast a DHCPOFFER packet
3. DHCP client broadcasts a DHCPREQUEST packet
4. DHCP Server1 broadcasts a DHCPACK (NAK) packet

4 steps in DHCP Lease Generation

1. The DHCP client broadcasts a DHCPDISCOVER packet.
2. Any DHCP Server in the subnet will respond by broadcasting a DHCPOFFER packet.
3. The client receives the DHCPOFFER packet.
4. The DHCP servers receive the DHCPREQUEST.

How DHCP Lease Renewal Works

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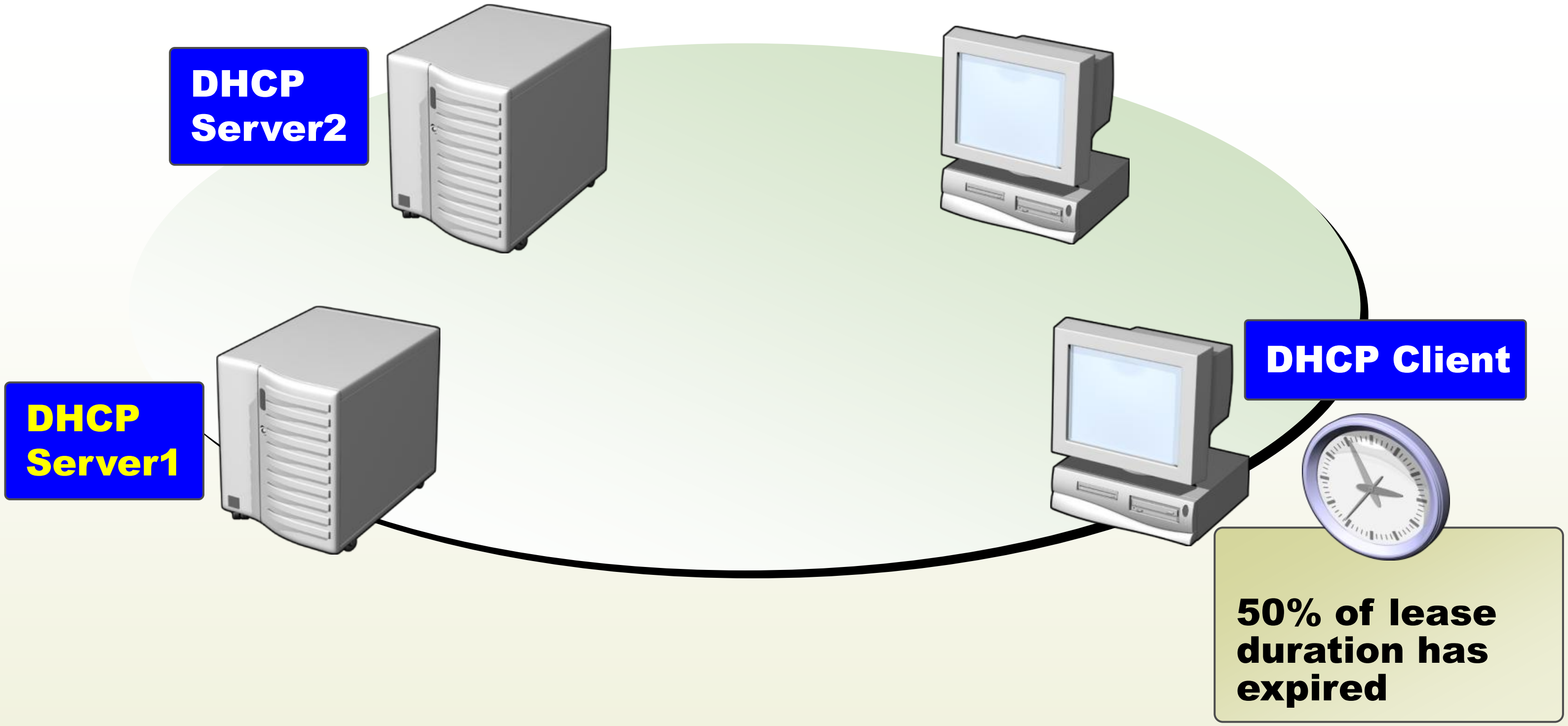

1. DHCP client sends a DHCPREQUEST packet
2. DHCP Server1 sends a DHCPACK packet

What Are DHCP Scopes?

scope = range of IP addresses for leasing

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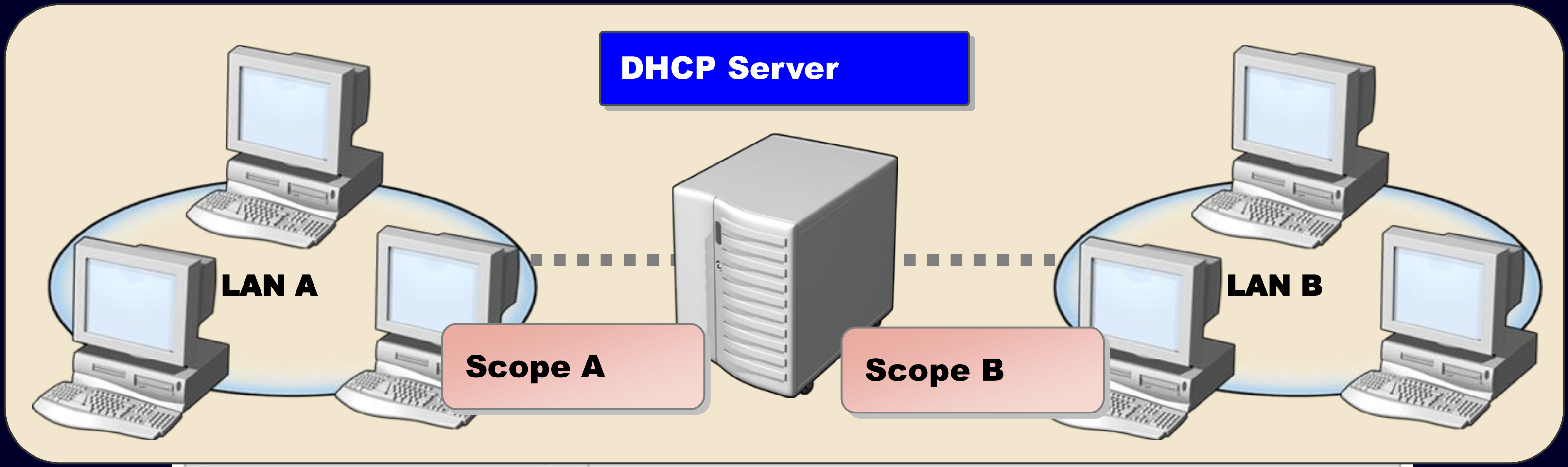

Scope Property	
Network ID	The network ID for the range of IP addresses
Subnet mask	The subnet mask for the network ID
Network IP address range	The range of IP addresses that are available to clients
Lease duration	The period of time that the DHCP server holds a leased IP address for a client before removing the lease.
Router	A DHCP option that allows DHCP clients to accessremote networks.
Scope name	An alphanumeric identifier for administrative purposes.
Exclusion range	The range of IP addresses in the scope that is excluded from being leased.

What Is a DHCP Reservation?

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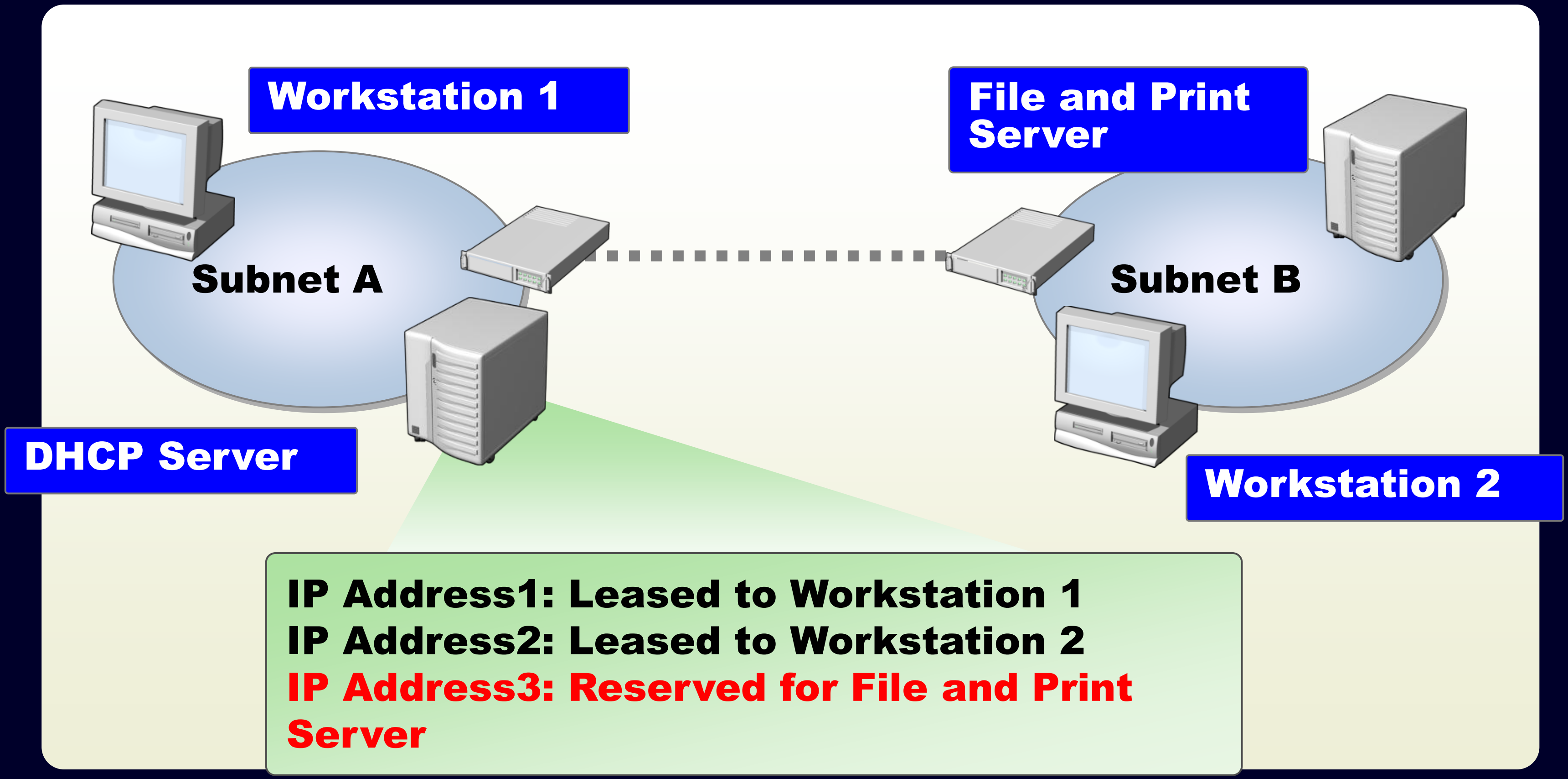
Reservation = fixed IP address for specific DHCP client



Implement DHCP Reservation

- Servers or printers is desirable to be with a fixed address (a guaranteed IP address). This is to avoid assigning inadvertently to another device.
- To configure a reservation, Server must obtain the device's MAC address.

```
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.1  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::20c:29ff:fead:9ec8  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:ad:9e:c8  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2522  bytes 265313 (259.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::4915:d27c:ca43:d423%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.188.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Wednesday, 29 July 2026 7:36:36 PM
   Lease Expires . . . . . . . . . . : Wednesday, 12 August 2026 10:36:51 AM
   Default Gateway . . . . . . . . . :
```

DHCP Sizing and Availability

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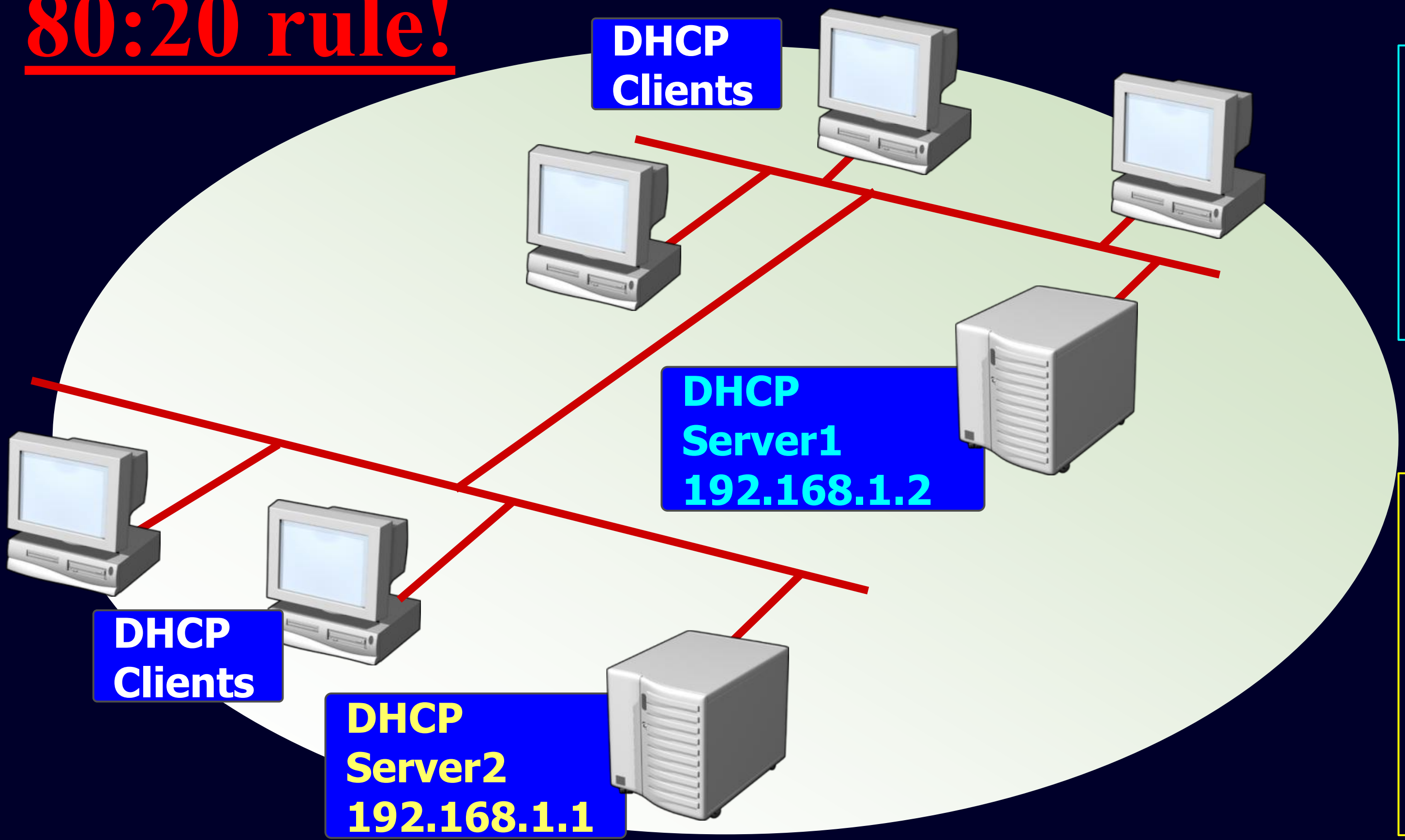
80:20 rule!



DHCP Server1

- 80% of addresses:
- 192.168.1.60 - 192.168.1.254

DHCP Server2

- 20% of addresses:
- 192.168.1.10 - 192.168.1.59

- Scope: 192.168.1.10 – .254
  (reserve .1 - .9 for servers)

The fault tolerance to the DHCP servers by increasing their availability.

scope = range of IP addresses for leasing

DHCP Server

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- Package: `kea` (obsolete: `dhcp-server, dhclient` and `dhcpd`)
  - **`dnf install kea`** # install the DHCP Server Package on your system
  - **`systemctl status kea-dhcp4`** # checks the status of the DHCP server (IPv4)
- Configuration file: **`/etc/kea/kea-dhcp4.conf`**
  Required settings: interface, default router, DNS, domain-search and optional: log server, hostname, requested ip address, TFTP server, lease time, etc.

```
"Dhcp4": {
    "interfaces-config": {
        "interfaces": ["ens192" ]
    },
    "control-socket": {
        "socket-type": "unix",
        "socket-name": "kea4-ctrl-socket"
    },
    "lease-database": {
        "type": "memfile",
        "lfc-interval": 3600
    },
    "expired-leases-processing": {
        "reclaim-timer-wait-time": 10,
        "flush-reclaimed-timer-wait-time": 25,
        "hold-reclaimed-time": 3600,
        "max-reclaim-leases": 100,
        "max-reclaim-time": 250,
        "unwarned-reclaim-cycles": 5
    },
    "renew-timer": 900,
    "rebind-timer": 1800,
    "valid-lifetime": 3600,
```

```
"option-data": [
    {
        "name": "domain-name-servers",
        "data": "8.8.8.8"
    },
    {
        "code": 15,
        "data": "example.org"
    },
    {
        "name": "domain-search",
        "data": "it.uts.edu.au"
    },
    {
        "name": "boot-file-name",
        "data": "EST5EDT4\\,M3.2.0/02:00\\,M11.1.0/02:00"
    },
    {
        "name": "default-ip-ttl",
        "data": "0xf0"
    }
],
```

DHCP Server (Ext.)

- Individual subnet configuration: `/etc/kea/kea-dhcp4.conf`

```
"subnet4": [
    {
        "id": 1,
        "subnet": "10.0.2.0/24",
        "pools": [ { "pool": "10.0.2.101 - 10.0.2.200" } ],
        "option-data": [
            {
                "name": "routers",
                "data": "10.0.2.1"
            }
        ],
        "reservations": [
            {
                "hw-address": "1a:1b:1c:1d:1e:1f",
                "ip-address": "10.0.2.254"
            }
        ]
    }
],
```

DHCP Server (Ext.)

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- Start servers and check the DHCP settings
  - `systemctl {start|status} kea-dhcp4`
  - Firewall configuration:
    - `firewall-cmd --add-service=dhcp --permanent`
    - `firewall-cmd --reload`
  - `tail /var/lib/kea/kea-leases4.csv` #check the lease log
  - Windows as a client:
    - `ipconfig /release`
    - `ipconfig /renew`
    - `ipconfig /all`

Networking Skills Test Assessment 1

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- Date and Time
  - Lab session in Week 5
- How to do
  - Questions: basic networking setting (static + dynamic)
  - Provide screenshots to answer the questions.
  - Use blank VMs: CentOS 10 and Windows Server 2025, or check out your previous snapshot

Note:

1. You need to use a portable hard drive (SSD or USB3 flash drive, with 64GB in minimal) to import the two VMs.
2. Format your hard drive to exFAT. Do not use NTFS or other file systems.
3. You need to download the empty *.ova files and import them into *.vmx before attending the skill test in Week 5.
4. Bring your student card when attending the skill test.
5. Please read the announcement given by the course coordinator.